

PDF → Multimodal Retrieval Index

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1. Objective

Turn an arbitrary PDF into a list of retrieval-ready chunks that preserve three different kinds of content — prose, tables, and figures — then embed those chunks and make them searchable.

Requirements as specified:

#	Requirement
R1	Partition with <code>unstructured</code> , <code>hi_res</code> strategy, extracting tables and images
R2	Chunk text by title: min 1000, soft 3000, hard 5000 characters
R3	~500 characters of overlap between chunks
R4	Page number and section title in every chunk's metadata
R5	Images: base64 in metadata, captioned by a local model, OCR keywords (stopwords removed) appended
R6	Tables: converted to markdown; caption first; split with headers repeated if over 5000 chars
R7	Embed with a small (<400M) current model, store in ChromaDB

All seven are implemented. Section 6 reports which are verified against real data and which are only covered by synthetic tests.

2. Test document

File	<code>relevant_section_identification-sample.pdf</code>
Content	Analog Devices / Maxim MAX77751 3.15A USB-C autonomous charger datasheet
Pages	40
Size	1,819,441 bytes
SHA-256	<code>0C7121C956C97EC5CFB809D388631D1473B8208A25DD4F84FE3E790AB57471B1</code>

Verified byte-identical to `AI Projects/relevant-section-identification/data/sample.pdf` .

A datasheet is a deliberately hard case: dense multi-column specification tables, dot-leader rows, oscilloscope traces, mechanical drawings, and rotated axis labels.

3. Environment

Component	Version
Python	3.11.9 (venv at C:\Documents2\resume-optimizer\.venv)
unstructured / unstructured-inference	0.25.2 / 1.6.13
transformers / sentence-transformers	5.9.0 / 5.5.1
torch	2.13.0+ cu126
chromadb	1.5.9
Tesseract OCR	5.4.0
Poppler	25.07
GPU	NVIDIA RTX 3050 Laptop, 4 GB VRAM , driver 555.97

Two environment issues worth recording:

- **CUDA wheel selection.** `pip install --upgrade torch --index-url .../cu124` silently no-ops when the index's newest torch is older than the installed CPU build. The exact variant must be pinned: `torch==2.13.0+cu126`. CUDA 13 wheels need a 580+ driver and are unusable here.
- **Binary discovery.** Tesseract was installed but absent from `PATH`. Since both `pytesseract` and `unstructured`'s OCR shell out to the binary, the pipeline patches `PATH` in-process at import (`ensure_binaries_on_path()`) rather than requiring a system change.

4. Architecture

PDF

```
└ partition_pdf(strategy="hi_res", infer_table_structure=True,
    extract_image_block_to_payload=True) → 852 elements
  ├── text elements (787) → chunk_by_title(1000/3000/5000, overlap=500) → 53 text chunks
  ├── Table elements (19) → HTML → markdown, caption first, split+repeat → 19 table chunks
  └ Image elements (46) → LLaVA caption + Tesseract OCR → keywords → 46 image chunks
      |
      | sort by page, apply cross-chunk overlap ▼
      | 118 chunks (JSON)
      |
      | EmbeddingGemma-300m, 768-dim, cosine ▼
      | ChromaDB collection
```

4.1 Text

`chunk_by_title` with `combine_text_under_n_chars=1000`, `new_after_n_chars=3000`, `max_characters=5000`, `overlap=500`, `overlap_all=True`.

`overlap_all=True` is the load-bearing flag: without it `unstructured` applies overlap only when splitting an oversized element, not between adjacent chunks.

The section title is recovered from each chunk's `orig_elements`, falling back to the last-seen `Title` so chunks that continue a section inherit its heading.

4.2 Tables

`infer_table_structure=True` populates `metadata.text_as_html`, converted to markdown with BeautifulSoup. Row 0 becomes the header, all rows are padded to the widest row, and literal pipes are escaped as `\|`.

Captions are found by scanning up to two elements either side of the table on the same page for a `FigureCaption` or text matching `^(table|tbl|exhibit)`.

Splitting: when a table exceeds 5000 characters it is divided on row boundaries, each part re-emitting the caption and the header + separator rows, and carrying the trailing rows of the previous part (up to ~500 chars) as overlap. Splits land on row boundaries, so overlap rounds *down* to whole rows — never mid-row, which would corrupt the markdown.

4.3 Images

Base64 comes straight from `unstructured`'s payload. Each image is captioned by a local LLaVA model and OCR'd with Tesseract; OCR text is tokenised, stopwords removed (NLTK, with a built-in fallback list), and the keywords appended to the caption. The chunk text is `caption + "\n\nKeywords: " + keywords`; the base64 lives in `metadata.image_base64`.

Model choice. The default is `llava-hf/llava-1.5-7b-hf`, but 4 GB of VRAM cannot hold it even in 4-bit once the vision tower and activations are resident. This run used `llava-hf/llava-interleave-qwen-0.5b-hf`. That constraint drives the single largest quality limitation in this report (§7.1).

4.4 Cross-chunk overlap

`chunk_by_title` handles text→text overlap. Table splits carry their own row overlap. A final pass gives the remaining chunks — every image chunk and the first part of every table — a 500-character prefix taken from the preceding chunk, preserving the un-prefixed version in `metadata.core_text`.

4.5 Embedding and storage

`google/embeddinggemma-300m` (308M parameters, 768 dimensions, 2048-token window). The model is prompt-conditioned and **asymmetric**:

- documents: `title: {title} | text: {content}`
- queries: `task: search result | query: {q}`

The checkpoint's default document prompt fills the title slot with the literal string `none`. Since every chunk carries a real section title, the indexer substitutes it — that slot exists for exactly this purpose.

Vectors are L2-normalised and stored in a persistent ChromaDB collection with `hnsw:space=cosine`. Chroma metadata accepts only scalars, so `ocr_keywords` is flattened to a comma-joined string.

The model is gated on HuggingFace; the licence must be accepted and the machine authenticated (`hf auth login`) before the weights will download.

5. Results

5.1 Chunk inventory

Type	Count	Core chars (excl. overlap)
text	53	1004 – 2177
table	19	55 – 2130
image	46	362 – 916
Total	118	output <code>sample_full.json</code> , 5.32 MB

Pages 1–40, all 40 distinct pages represented.

5.2 Runtime

Stage	Time
<code>hi_res</code> partition (40 pages, CPU)	~15 min — cached thereafter via <code>--cache-dir</code>
Captioning 46 images (RTX 3050)	~4 min
Captioning, CPU, for comparison	193 s for a single image
Embedding 118 chunks	~15 s

The layout model runs through `onnxruntime` (CPU build), so partitioning does not benefit from the GPU; only captioning does.

5.3 Retrieval evaluation

10 questions with hand-labelled answer pages:

Candidate pool	recall@1	recall@3	recall@5
All chunk types	70%	90%	90%
Text + tables only	80%	90%	90%

6. Verification

Measured, not asserted — every figure below comes from an automated check over the full output rather than inspection of samples.

Property	Method	Result
Hard size limit	all 118 chunks vs 5000 + 500	0 violations
Min-size rule	text chunks < 1000 chars	0
Text→text overlap	longest suffix/prefix match	52/52 pairs, 498–500 chars
Image/table overlap	prefix byte-compared to previous chunk's tail	65/65 exact
<code>core_text</code> integrity	reconstruct <code>text</code> minus prefix	65/65 exact
Markdown validity	header + separator + column counts	19/19 valid, 0 ragged rows
Page + title present	all chunks	118/118
Base64 decodes	all image chunks rendered in a browser	46/46, 0 failures
Chroma round-trip	re-read metadata after insert	base64, keywords, <code>core_text</code> intact

7. Limitations

Stated plainly; each is a real constraint on how far this output can be trusted.

7.1 Image captions are unreliable — and measurably costly

The 0.5B captioner hallucinates on technical figures. The page-23 charger state machine is described as *"a green circle labeled 'Input' and a red circle labeled 'Output'"*; no such circles exist.

This is not cosmetic. For the query *"what resistor value sets the top-off current?"*, a pin-diagram image chunk captioned *"a black and white schematic diagram of a computer circuit board"* scored **cos 0.505**, outranking `Table 4. Top-Off Current Settings` — which contains the literal answer — at **cos 0.446**. Excluding image chunks raises recall@1 from 70% to 80%, and that 10-point gap is attributable to this single query flipping.

The fix is a larger captioner, which this GPU cannot host. Mitigations that do not require better hardware: lead image chunks with OCR keywords rather than the caption, and suppress captions for the 15 of 46 images where OCR returned nothing (those chunks are currently pure hallucination with no counterweight).

7.2 OCR quality on this document class

- Dot-leader rows read as runs of `e: CHGIN to GND 20.00... eee eeeee .`
- Rotated axis labels are garbled: `anoqd , soyvho , jyvlsay , lroporr .`
- Subscripts are lost: `V_BUS` → `Vpus` , `μA` → `UA` / `HA` .
- **15 of 46 images** produced no usable keywords at all.

7.3 Table structure

- **3 of 19 tables have a data row as their header**, because the source HTML places data inside `<thead>` .
Harmless at current sizes, but a table large enough to split would repeat that data row as the header of every part.
- **Caption detection reached only 8 of 19 tables**, and one of those 8 is wrong — a package-drawing note (`( 3mm x 3mm , 0.4mm PITCH )`) captured as a table caption.

7.4 The overlap/markdown conflict

Requirements R3 (overlap everywhere) and R6 (tables as valid markdown) conflict. In **5 table chunks** the borrowed 500 characters come from another table, so the chunk's raw `text` contains a truncated fragment of a

foreign pipe-table above its own. `core_text` is always clean; only the concatenated `text` is affected. This is an open design decision, not a defect — the candidate resolutions are to strip pipe rows from the borrowed tail, or to move the overlap into metadata for table chunks only.

7.5 Untested code paths

No table in this document exceeded 5000 characters and no text chunk exceeded the 3000-character soft limit. The table-splitting logic — header repetition and row-level overlap — is therefore verified **only against a synthetic 27,000-character table** (6 parts, each ≤ 5000 chars, ~ 408 chars of shared rows between consecutive parts), never against real input.

7.6 The evaluation is self-authored

Both the questions and the gold page labels in §5.3 were written by the same person who had already read the chunks while debugging. They are a sanity check, not an unbiased benchmark, and are biased toward content known to be present. Ten cases is also too few for the percentages to carry meaningful confidence intervals.

8. Reproduction

```
pip install -r requirements.txt
```

System binaries required: `tesseract`, `poppler`. GPU users must pin an explicit CUDA torch build (see `SETUP.md`).

```
hf auth login          # EmbeddingGemma is gated
```

```
python pdf_chunker.py sample.pdf --model llava-hf/llava-interleave-qwen-0.5b-hf --device cuda --cache-dir .cache --out chunks.json
```

```
python embed_index.py chunks.json --db ./chroma --collection max77751 --device cuda --reset
```

```
python query_index.py "what does the STAT pin indicate?" -k 5
```

9. Files

File	Purpose
<code>pdf_chunker.py</code>	Partition → text/table/image chunks
<code>embed_index.py</code>	Embed with EmbeddingGemma → ChromaDB
<code>query_index.py</code>	Query the index (applies the query prompt)
<code>eval_retrieval.py</code>	Recall@k over labelled questions
<code>requirements.txt</code> , <code>SETUP.md</code>	Dependencies and setup
<code>sample_full.json</code>	118 chunks from the test document
<code>chroma/</code>	Persistent vector store

10. Next steps

1. Re-order image chunk text to lead with OCR keywords; drop captions where OCR found nothing. Cheapest available recall improvement (§7.1).
2. Resolve the overlap/markdown conflict for table chunks (§7.4).
3. Build a held-out evaluation set authored without sight of the chunks (§7.6).
4. Exercise table splitting against a document containing a table over 5000 characters (§7.5).
5. Detect header rows rather than assuming row 0 (§7.3).